

Emanuele Bugliarello

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Research Interests

Representation learning, natural language and vision understanding, multi-modal learning, transfer learning

Education

MSc in Communication Systems, Data Analytics specialization 2015–2018
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland

Thesis: *Matrix Completion in the Unit Hypercube via Structured Matrix Factorization*
GPA: 5.53/6 (3/44)

BEng in Electronics and Information Engineering 2013–2018
Tongji University (同济大学) Shanghai, China

Thesis: *General large-batch methods for scalable accurate training of neural networks*
GPA: 29.34/30

BSc in Telecommunications Engineering 2012–2015
Politecnico di Torino Turin, Italy

GPA: 28.86/30 (3/120) – Graduation Grade: 110/110 Cum Laude

Experience

Research Associate with Prof. Naoaki Okazaki 10/2018–present
Okazaki Lab @ Tokyo Institute of Technology (東京工業大学) Tokyo, Japan
On Neural Machine Translation for News Articles and Multi-Modal Neural Machine Translation

Visiting Graduate Student with Prof. Rui Fan and Prof. Dan Alistarh 04/2018–07/2018
ShanghaiTech University (上海科技大学) Shanghai, China
Large-Batch Algorithms for Deep Learning at Scale

- Implementation of state-of-the-art algorithms for large-batch training of Convolutional Neural Networks
- Evaluation of large-batch training algorithms performance on different datasets to assess their generality

Research Intern with Swayambhoo Jain, Brian Eriksson and Jean Bolot 09/2017–03/2018
Technicolor AI Lab Palo Alto, CA, USA
Intelligent VFX Scheduling

- Predicting visual effects (VFX) assets overspending by blending classification and censored regression models
- Predicting weekly number of required employees in VFX departments using advanced regression methods
- Recommending employees to VFX departments with novel structured matrix factorization models

Research Assistant with Marcel Schöngens, Maxime Martinasso and Claudio Gheller Summer 2017
Swiss National Supercomputing Centre (CSCS) Lugano, Switzerland
Distributed TensorFlow: A performance evaluation

- Deployment of distributed TensorFlow on the Piz Daint supercomputer and various AWS EC2 clusters
- Scalability study of TensorFlow using InceptionV3 for image classification on ImageNet

Research Assistant with Prof. Jean-Pierre Hubaux Summer 2016
Computer Communications and Applications Laboratory @ EPFL Lausanne, Switzerland
Quantifying the effect of Co-location information on Location Privacy on Online Social Networks

- Design and implementation of a Web scraper to collect Facebook users' public data
- Development of a parser to extract declared and inferable location and co-location data

Teaching

Teaching Assistant, EPFL: Internet Analytics (COM-308)

Spring 2017

Projects

Evaluating the evolution of Wikipedia's navigability

Spring 2017

Semester project with Prof. Robert West

Data Science Lab @ EPFL

Investigate Wikipedia's navigability over the course of its history, based on the English Wikipedia's complete edit history

- Design of a parser to extract hyperlinks (including hatnotes) from Wiki Markup Language
- Processing of a big data set (6 TB) consisting of all the revisions in the English Wikipedia with Spark
- Analysis of the temporal evolution of Wikipedia's graph properties
- Poster accepted at Wiki Workshop, WWW 2018

Solarity: Solar Clarity for Swiss Households

Fall 2016

Applied Data Analysis (CS-401)

EPFL

Based on historical weather data, predict solar energy production in the coming years to provide information on initial investment, break-even time, and potential savings for domestic solar installations

- Development of a Web scraper to retrieve publicly available weather data
- Predicting solar potential in the coming years with a regression model based on historic data
- Design of a responsive website with a clean interface and that is easily usable
- 2nd best class project award (over 60+ projects)

Twitter Sentiment Classification

Fall 2016

Machine Learning (CS-433)

EPFL

Examine the accuracy of different learning algorithms in determining sentiments associated to tweets

- Text preprocessing: replace repeating letters, stemming, n-grams, remove stop words
- Word embeddings: TF-IDF, GloVe
- Classifiers: Logistic Regression, multinomial naive Bayes, SVM with linear kernel

SecureRide: A cheat-proof ride-hailing service

Fall 2016

Advanced Topics on Privacy Protection (CS-622)

EPFL

Mitigate the integrity threat due to GPS manipulation by a dishonest driver to over-charge riders

- Evaluating lower-bound distance of a trip via secure-location techniques (Location Proofs)
- Achievement of tight bounds by making use of street-map networks

Skills

Programming

- Python, C, Java, Matlab, Java for Android
- NumPy, Pandas, scikit-learn, CUDA
- PyTorch, TensorFlow
- Spark, Hadoop MapReduce, MPI
- SQL, MySQL, Mongo

Languages

- Italian (native)
- English (fluent – TOEFL iBT 109)
- French (intermediate)
- Chinese (intermediate – HSK III)
- Japanese (basic)

Fellowships, Honors, and Awards

EPFL PhD Program in Computer Science Fellowship [offer declined]

01/2018

EPFL TCP/IP 2015-2016 Research Exercise Award

12/2015

EIT Digital Master School Excellence Scholarship [offer declined]

04/2015

PoliTong Merit Scholarship

07/2013

Politecnico di Torino fees waiver for academic merit

2013–2015

MIUR National Register of Excellences

06/2012